

CHAPTER

7.6

DIGITAL TRANSMISSION

Statement for Question 1-5

Fig. P7.6.1-5 shows fourier spectra of signal $x(t)$ and $y(t)$. Determine the Nyquist sampling rate for the given function in question.

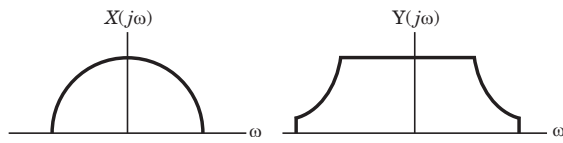


Fig.P7.6.1-5

1. $x(t)$

- (A) 100 kHz (B) 200 kHz
(C) 300 kHz (D) 50 kHz

2. $y(t)$

- (A) 50 kHz (B) 75 kHz
(C) 150 kHz (D) 300 kHz

3. $x^2(t)$

- (A) 100 kHz (B) 150 kHz
(C) 250 kHz (D) 400 kHz

4. $y^3(t)$

- (A) 100 kHz (B) 300 kHz
(C) 900 kHz (D) 120 kHz

5. $x(t)y(t)$

- (A) 250 kHz (B) 500 kHz
(C) 50 kHz (D) 100 kHz

Statement for Question 6-7

A signal $x(t)$ is multiplied by rectangular pulse train $c(t)$ shown in fig.P7.6.6-7..

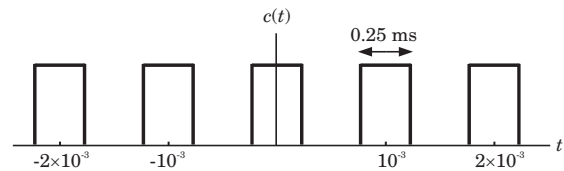


Fig.P7.6.6-7

6. $x(t)$ would be recovered from the product. $x(t)c(t)$ by using an ideal LPF if $X(j\omega) = 0$ for

- (A) $\omega > 2000\pi$ (B) $\omega > 1000\pi$
(C) $\omega < 1000\pi$ (D) $\omega < 2000\pi$

7. If $X(j\omega)$ satisfies the constraints required, then the pass band gain A of the ideal lowpass filter needed to recover $x(t)$ from $e(t)x(t)$ is

- (A) 1 (B) 2
(C) 4 (D) 8

8. Consider a set of 10 signals $x_i(t)$, $i = 1, 2, 3, \dots, 10$. Each signal is band limited to 1 kHz. All 10 signals are to be time-division multiplexed after each is multiplied by a carrier $e(t)$ shown in Figure. If the period T of $e(t)$ is chosen to have the maximum allowable value, the largest value of Δ would be

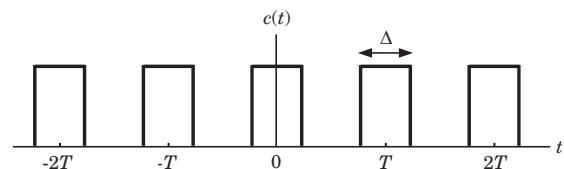


Fig.P7.6.8

- (A) 5×10^{-3} sec (B) 5×10^{-4} sec
(C) 5×10^{-5} sec (D) 5×10^{-6} sec

9. A compact disc recording system samples a signals with a 16-bit analog-to-digital convertor at 44.1 kbits/s. The CD can record an hours worth of music. The approximate capacity of CD is

- (A) 705.6 M Bytes (B) 317.5 M Bytes
(C) 2.54 M Bytes (D) 5.43 M Bytes

10. An analog signal is sampled at 36 kHz and quantized into 256 levels. The time duration of a bit of the binary coded signal is

- (A) 5.78 μ s (B) 3.47 μ s
(C) 6.43 ms (D) 7.86 ms

11. An analog signal is quantized and transmitted using a PCM system. The tolerable error in sample amplitude is 0.5% of the peak-to-peak full scale value. The minimum binary digits required to encode a sample is

- (A) 5 (B) 6
(C) 7 (D) 8

Statement for Question 12-13.

Ten telemetry signals, each of bandwidth 2kHz, are to be transmitted simultaneously by binary PCM. The maximum tolerable error in sample amplitudes is 0.2% of the peak signal amplitude. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 1% extra bits.

12. The minimum possible data rate must be

- (A) 272.64 k bits/sec (B) 436.32 k bits/sec
(C) 936.32 k bits/sec (D) None of the above

13. The minimum transmission bandwidth is

- (A) 218.16 kHz (B) 468.32 kHz
(C) 136.32 kHz (D) None of the above

14. A Television signal is sampled at a rate of 20% above the Nyquist rate. The signal has a bandwidth of 6 MHz. The samples are quantized into 1024 levels. The minimum bandwidth required to transmit this signal would be

- (A) 72 M bits/sec (B) 144 M bits/sec
(C) 72 k bits/sec (D) 144 k bits/sec

15. A CD record audio signals digitally using PCM. The audio signal bandwidth is 15 kHz. The Nyquist samples are quantized into 32678 levels and then binary coded. The minimum number of binary digits required to encode the audio signal

- (A) 450 k bits/sec (B) 900 k bits/sec
(C) 980 340 k bits/sec (D) 490 170, k bits/sec

16. The American Standard Code for Information Interchange has 128 characters, which are binary coded. If a certain computer generates 1,000,000 character per second, the minimum bandwidth required to transmit this signal will be

- (A) 1.4 M bits/sec (B) 14 M bits/sec
(C) 7 M bits/sec (D) 0.7 M bits/sec

17. A binary channel with capacity 36 k bits/sec is available for PCM voic transmission. If signal is band limited to 3.2 kHz, then the appropriate values of quantizing level L and the sampling frequency will be

- (A) 32, 3.6 kHz (B) 64, 7.2 kHz
(C) 64, 3.6 kHz (D) 32, 7.2 kHz

18. Fig.P7.4.18 shows a PCM signals in which amplitude level of + 1 volt and - 1 volt are used to represent binary symbol 1 and 0 respectively. The code word used consists of three bits. The sampled version of analog signal from which this PCM signal is derived is

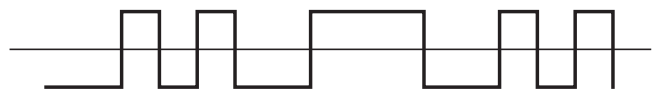


Fig.P7.4.18

- (A) 4 5 1 2 1 3 (B) 8 4 3 1 2
(C) 6 4 3 1 7 (D) 1 2 3 4 5

19. A PCM system uses a uniform quantizer followed by a 8-bit encoder. The bit rate of the system is equal to 10^8 bits/s. The maximum message bandwidth for which the system operates satisfactorily is

- (A) 25 MHz (B) 6.25 MHz
(C) 12.5 MHz (D) 50 MHz

20. Twenty-four voice signals are sampled uniformly at a rate of 8 kHz and then time-division multiplexed. The sampling process uses flat-top samples with 1 μ s duration. The multiplexing operating includes provision

for synchronization by adding an extra pulse of 1 μ s duration. The spacing between successive pulses of the multiplexed signal is

- (A) 4 μ s (B) 6 μ s
(C) 7.2 μ s (D) 8.4 μ s

21. A linear delta modulator is designed to operate on speech signals limited to 3.4 kHz. The sampling rate is 10 times the Nyquist rate of the speech signal. The step size δ is 100 mV. The modulator is tested with a test signal required to avoid slope overload is

- (A) 2.04 V (B) 1.08 V
(C) 4.08 V (D) 2.16 V

Statement for Question 22-23 :

Consider a linear DM system designed to accommodate analog message signals limited to a bandwidth of 3.5 kHz. A sinusoidal test signal of amplitude $A_{\max} = 1$ V and frequency $f_m = 800$ Hz is applied to the system. The sampling rate of the system is 64 kHz.

22. The minimum value of the step size to avoid overload is

- (A) 240 mV (B) 120 mV
(C) 670 mV (D) 78.5 mV

23. The granular-noise power would be

- (A) 1.68×10^{-3} W (B) 2.86×10^{-4} W
(C) 2.48×10^{-3} W (D) 1.12×10^{-4} W

24. The SNR will be

- (A) 298 (B) 1.75×10^{-3}
(C) 4.46×10^3 (D) 201

25. The output signal-to-quantization-noise ratio of a 10-bit PCM was found to be 30 dB. The desired SNR is 42 dB. It can be increased by increasing the number of quantization levels. In this way the fractional increase in the transmission bandwidth would be (assume $\log_2 10 = 0.3$)

- (A) 20% (B) 30%
(C) 40% (D) 50%

Statement for Question 26-27.

A signal has a bandwidth of 1 MHz. It is sampled at a rate 50% higher than the Nyquist rate and

quantized into 256 levels using a μ -law quantizer with $\mu = 225$.

26. The signal-to-quantization-noise ratio is

- (A) 34.91 dB (B) 38.06 dB
(C) 42.05 dB (D) 48.76 dB

27. It was found that a sampling rate 20% above the rate would be adequate. So the maximum SNR, that can be realized without increasing the transmission bandwidth, would be

- (A) 60.4 dB (B) 70.3 dB
(C) 50.1 dB (D) None of the above

28. For a PCM signal the compression parameter $\mu = 100$ and the minimum signal to quantization-noise ratio is 50 dB. The number of bits per sample would be.

- (A) 8 (B) 10
(C) 12 (D) 14

29. A sinusoidal message signal $m(t)$ is transmitted by binary PCM without compression. If the signal-to-quantization-noise ratio is required to be at least 48 dB, the minimum number of bits per sample will be

- (A) 8 (B) 10
(C) 12 (D) 14

30. A speech signal has a total duration of 20 sec. It is sampled at the rate of 8 kHz and then PCM encoded. The signal-to-quantization noise ratio is required to be 40 dB. The minimum storage capacity needed to accommodate this signal is

- (A) 1.12 KBytes (B) 140 KBytes
(C) 168 KBytes (D) None of the above

31. The input to a linear delta modulator having a step-size $\Delta = 0.628$ is a sine wave with frequency f_m and peak amplitude E_m . If the sampling frequency $f_s = 40$ kHz, the combination of the sine-wave frequency and the peak amplitude, where slope overload will take place is

- | | |
|-----------|-------|
| E_m | f_m |
| (A) 0.3 V | 8 kHz |
| (B) 1.5 V | 4 kHz |
| (C) 1.5 V | 2 kHz |
| (D) 3.0 V | 1 kHz |

32. A sinusoidal signal with peak-to-peak amplitude of 1.536 V is quantized into 128 levels using a mid-rise uniform quantizer. The quantization-noise power is

- (A) 0.768 V (B) $48 \times 10^{-6} V^2$
(C) $12 \times 10^{-6} V^2$ (D) 3.072 V

33. A signal is sampled at 8 kHz and is quantized using 8 bit uniform quantizer. Assuming SNR_q for a sinusoidal signal, the correct statement for PCM signal with a bit rate of R is

- (A) $R = 32$ kbps, $SNR_q = 25.8$ dB
(B) $R = 64$ kbps, $SNR_q = 49.8$ dB
(C) $R = 64$ kbps, $SNR_q = 55.8$ dB
(D) $R = 32$ kbps, $SNR_q = 49.8$ dB

34. A 1.0 kHz signal is flat-top sampled at the rate of 180 samples/sec and the samples are applied to an ideal rectangular LPF with cut-off frequency of 1100 Hz, then the output of the filter contains

- (A) only 800 Hz component
(B) 800 and 900 Hz components
(C) 800 Hz and 1000 Hz components
(D) 800 Hz, 900 and 1000 Hz components

35. The Nyquist sampling interval, for the signal $\text{sinc}(700t) + \text{sinc}(500t)$ is

- (A) $\frac{1}{350}$ sec (B) $\frac{\pi}{350}$ sec
(C) $\frac{1}{700}$ sec (D) $\frac{\pi}{175}$ sec

36. A signal $x(t) = 100 \cos(24\pi \times 10^3)t$ is ideally sampled with a sampling period of 50 μ sec and then passed through an ideal lowpass filter with cutoff frequency of 15 KHz. Which of the following frequencies is/are present at the filter output

- (A) 12 KHz only (B) 8 KHz only
(C) 12 KHz and 9 KHz (D) 12 KHz and 8 KHz

37. In a PCM system, if the code word length is increased from 6 to 8 bits, the signal to quantization noise ratio improves by the factor.

- (A) 8/6 (B) 12
(C) 16 (D) 8

38. Four signals $g_1(t)$, $g_2(t)$, $g_3(t)$ and $g_4(t)$ are to be multiplexed and transmitted. $g_1(t)$ and $g_4(t)$ have a bandwidth of 4 kHz, and the remaining two signals have bandwidth of 8 kHz. Each sample requires 8 bit for encoding. What is the minimum transmission bit rate of the system.

- (A) 512 kbps (B) 16 kbps
(C) 192 kbps (D) 384 kbps

39. Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz, are sampled at their respective Nyquist rates, encoded with 12 bit words, and time division multiplexed. The bit rate for the multiplexed signal is

- (A) 115.2 kbps (B) 28.8 kbps
(C) 57.6 kbps (D) 38.4 kbps

40. The minimum sampling frequency (in samples/sec) required to reconstruct the following signal from its samples without distortion would be

- $$x(t) = 5 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^3 + 7 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^2$$

(A) 2×10^3 (B) 4×10^3
(C) 6×10^3 (D) 8×10^3

41. The minimum step-size required for a Delta-Modulator operating at 32 K samples/sec to track the signal (here $u(t)$ is the unit function)

$x(t) = 125t(u(t) - u(t-1)) + (250 - 125t)(u(t-1) - u(t-2))$ so that slope overload is avoided, would be

- (A) 2^{-10} (B) 2^{-8}
(C) 2^{-6} (D) 2^{-4}

42. Four signals each band limited to 5 kHz are sampled at twice the Nyquist rate. The resulting PAM samples are transmitted over a single channel after time division multiplexing. The theoretical minimum transmission bandwidth of the channel should be equal to.

- (A) 5 kHz (B) 20 kHz
(C) 40 kHz (D) 80 kHz

43. Four independent messages have bandwidths of 100 Hz, 100 Hz, 200 Hz and 400 Hz respectively. Each is

sampled at the Nyquist rate, time division multiplexed and transmitted. The transmitted sample rate, in Hz, is given by

- (A) 200 (B) 400
(C) 800 (D) 1600

44. The Nyquist sampling rate for the signal $g(t) = 10 \cos(50\pi t) \cos^2(150\pi t)$. Where 't' is in seconds, is

- (A) 150 samples per second
(B) 200 samples per second
(C) 300 samples per second
(D) 350 samples per second

45. A TDM link has 20 signal channels and each channel is sampled 8000 times/sec. Each sample is represented by seven binary bits and contains an additional bit for synchronization. The total bit rate for the TDM link is

- (A) 1180 K bits/sec (B) 1280 K bits/sec
(C) 1180 M bits/sec (D) 1280 M bits/sec

46. In a CD player, the sampling rate is 44.1 kHz and the samples are quantized using a 16-bit/sample quantizer. The resulting number of bits for a piece of music with a duration of 50 minutes is

- (A) 1.39×10^9 (B) 4.23×10^9
(C) 8.46×10^9 (D) 12.23×10^9

47. Four voice signals, each limited to 4 kHz and sampled at Nyquist rate are converted into binary PCM signal using 256 quantization levels. The bit transmission rate for the time-division multiplexed signal will be

- (A) 8 kbps (B) 64 kbps
(C) 256 kbps (D) 512 kbps

48. Analog data having highest harmonic at 30 kHz generated by a sensor has been digitized using 6 level PCM. What will be the rate of digital signal generated?

- (A) 120 kbps (B) 200 kbps
(C) 240 kbps (D) 180 kbps

49. In a PCM system, the number of quantization levels is 16 and the maximum signal frequency is 4 kHz.; the bit transmission rate is

- (A) 32 bits/s (B) 16 bits/s
(C) 32 kbits/s (D) 64 dbits/s

50. A speech signal occupying the bandwidth of 300 Hz to 3 kHz is converted into PCM format for use in digital communication. If the sampling frequency is 8 kHz and each sample is quantized into 256 levels, then the output bit rate will be

- (A) 3 kb/s (B) 8 kb/s
(C) 64 kb/s (D) 256 kb/s

51. If the number of bits in a PCM system is increased from n to $n + 1$, the signal-to-quantization noise ratio will increase by a factor.

- (A) $\frac{(n+1)}{n}$ (B) $\frac{(n+1)^2}{n^2}$
(C) 2 (D) 4

52. In PCM system, if the quantization levels are increased from 2 to 8, the relative bandwidth requirement will.

- (A) remain same (B) be doubled
(C) be tripled (D) become four times

53. Assuming that the signal is quantized to satisfy the condition of previous question and assuming the approximate bandwidth of the signal is W . The minimum required bandwidth for transmission of a binary PCM signal based on this quantization scheme will be.

- (A) 5 W (B) 10 W
(C) 20 W (D) None of the above

16. (C) $128=2^7$, We need 7 bits/character. For 1,000,000 character we need 7 Mbits/second. Thus minimum bandwidth = 7 Mbits/sec.

17. (D) $f_s > 2f_m = 6400$ Hz, $nf_s \leq 63000$
 $n \leq \frac{36000}{6400} = 5.63$, $n = 5$, $L = 2^n = 32$, $f_s = \frac{36000}{5} = 7.2$ kHz.

18. (D) The transmitted code word are

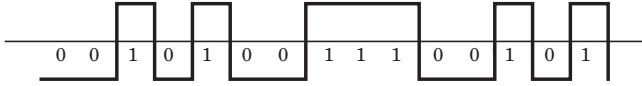


Fig.S.7.6.18

In 1st word 001(1)

In 2nd word 010(2)

In 3rd word 011(3)

In 4th word 100(4)

In 5th word 101 (5)

19. (B) Message bandwidth = W , Nyquist rate = $2W$

Bandwidth = $2W \times 8 = 16W$ bit/s

$16W = 10^8$, or $W = \frac{10^8}{16} = 6.25$ MHz

20. (A) Sampling interval $T_s = \frac{1}{8k} = 125 \mu s$. There are 24 channels and 1 sync pulse, so the time allotted to each channel is $T_c = \frac{T_s}{25} = 5 \mu s$. The pulse duration is $1 \mu s$. So the time between pulse is $4 \mu s$.

21. (B) $A_{\max} = \frac{\delta f_s}{\omega_m} = \frac{0.1 \times 68k}{2\pi \times 10^3} = 1.08$ V

22. (D) $A_{\max} = \frac{\delta f_s}{\omega_m}$ or $\delta = \frac{A_{\max} \omega_m}{f_s} = \frac{1 \times 2\pi \times 800}{50 \times 10^3} = 78.5$ mV

23. (D) $N_o = \frac{\delta^2 B}{3f_s} = \frac{(0.0785)^2 \times 3500}{3 \times 64000} = 1.122 \times 10^{-4}$ W

24. (C) $\frac{S_o}{N_o} = \frac{0.5}{1.12 \times 10^{-4}} = 4.46 \times 10^3$

25. (A) $\frac{S_o}{N_o} \propto L^2$, $L = 2^n \left(\frac{S_o}{N_o} \right)_{dB} = 10 \log(C2^{2n})$

$= \log C + 20n \log 2 = \alpha + 6n$ dB. This equation shows that increasing n by one bits increase the by 6 dB. Hence an increase in the SNR by 12 dB can be accomplished by increasing n from 10 to 12, the transmission bandwidth would be increased by 20%

26. (B) $\frac{S_o}{N_o} = \frac{3L^2}{[\ln(\mu + 1)]^2} = 6394 = 38.06$ dB

27. (C) Nyquist Rate = 2 MHz

50% higher rate = 3 MHz, $L = 256 = 2^8$

Thus transmission bandwidth is $3 \text{ MHz} \times 8 = 24$ Mbits/s.

New sampling rate is at 20% above the Nyquist rate.

Sampling rate = $1.2 \times 2 = 2.4$ MHz.

bits per second = $\frac{24M_{\text{sec}}^{\text{bits}}}{2.4 \text{ MHz}} = 10$ bits

Level = $2^{10} = 1024$, $\frac{S_o}{N_o} = \frac{3(1024)^2}{(\ln 256)^2} = 102300 = 50.1$ dB

28. (B) $\frac{S_o}{N_o} = \frac{3L^2}{[\ln(\mu + 1)]^2} \geq 50$ dB, $\mu = 100$

$\frac{3L^2}{[\ln 101]^2} = 100000$ or $L = 842.6$

Because L is power of 2, we select $L = 1024 = 2^{10}$.

Thus 10 bits are required.

29. (A) $S_o = \overline{m^2(t)}$, $N_o = \frac{3m_p^2}{3L^2}$, $\frac{S_o}{N_o} = \frac{3L^2 m^2(t)}{m_p^2}$

since signal is sinusoidal $\frac{\overline{m^2(t)}}{m_p^2} = \frac{1}{2}$,

$\frac{3L^2}{2} = 48$ dB = 63096, $L = 205.09$

Since L is power of 2, so we select $L = 256$

Hence $256 = 2^8$, So 8 bits per sample is required .

30. (B) $(SNR)_q = 1.76 + 6.02(n) = 40$ dB, $n = 6.35$

We take the $n = 7$.

Capacity = $20 \times 8k \times 7 = 1.12$ Mbits = 140 Kbytes

31. (B) For slope overload to take place $E_m \geq \frac{\delta f_s}{2\pi f_m}$

This is satisfied with $E_m = 1.5$ V and $f_m = 4$ kHz.

32. (C) Step size $\delta = \frac{2m_p}{L} = \frac{1536}{128} = 0.012$ V

quantization noise power

$= \frac{\delta^2}{12} = \frac{(0.012)^2}{12} = 12 \times 10^{-6}$ V²

33. (B) Bit Rate = $8k \times 8 = 64$ kbps

$(SNR)_q = 1.76 + 6.02n$ dB = $1.76 + 6.02 \times 8 = 49.8$ dB

34. (B) $f_s = 1800$ samples/sec, $f_m = \frac{1800}{2} = 900$ Hz

Since the sampling rate is 1800 samples/sec the highest frequency that can be recovered is 900 Hz.

35. (C) $x(t) = \text{sinc } 700t + \text{sinc } 500t$

$$= \frac{1}{\pi t} [\sin 700\pi t + \sin 500\pi t]$$

$x(t)$ is band limited with $f_m = 350$ Hz, Thus Nyquist rate is $2f_m = 700$ Hz, Sampling interval = $\frac{1}{700}$ sec

36. (D) $f_s = \frac{1}{T} = \frac{1}{50 \times 10^{-6}} = 20$ kHz, $f_c = 12$ kHz

The frequency passed through LPF are f_c , $f_s - f_m$ or 12 kHz, 8 kHz

37. (C) $P = \frac{(\text{SNR})_1}{(\text{SNR})_2} = \frac{2^{2n_2}}{2^{2n_1}}$, Here n = code word length,

$$n_1 = 6, n_2 = 8, \quad \text{Thus rate} = \frac{2^{16}}{2^{12}} = 16$$

38. (D) Signal $g_1(t)$, $g_2(t)$, $g_s(t)$ and $g_4(t)$ will have 8 k, 8 k, 16 k and 16 k sample/sec at Nyquist rate. Thus 48000 sample/sec bit rate $48000 \times 8 = 384$ kbps

39. (C) Analog signals, having bandwidth 1200 Hz, 600 Hz and 600 Hz have 2400, 1200 samples/sec at Nyquist rate. Hence 48000 sample/sec
bit rate = 48000 sample/sec $\times 12 = 57.6$ kbps

40. (C) $x(t) = 5 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^3 + 7 \left(\frac{\sin 2\pi 1000t}{\pi t} \right)^2$

Maximum frequency component = $3 \times 1000 = 3$ kHz

Sampling rate = $2f_m = 6$ kHz

41. (B) Here $f_s = 32$ k sample/sec

$$E_m = 125, \quad f_m = \frac{1}{T} = \frac{1}{2}$$

For slope-overload to be averted $E_m \geq \frac{f_s}{f_m}$

$$\Delta \leq \frac{E_m f_m}{f_s} \quad \text{or} \quad \Delta \leq \frac{125(\frac{1}{2})}{32 \times 10^3} \quad \text{or} \quad \Delta \leq \frac{(128)(\frac{1}{2})}{32 \times 1024} \quad \text{or} \quad \Delta \leq 2^{-8}$$

42. (D) $f_m = 5$ kHz, Nyquist Rate = $2 \times 5 = 10$ kHz
Since signal are sampled at twice the Nyquist rate so sampling rate = $2 \times 10 = 20$ kHz.
Total transmission bandwidth = $4 \times 20 = 80$ kHz

43. (D) Signal will be sampled 200, 200, 400 and 800 sample/sec thus 1600 sample per second,

44. (D) $g(t) = 10 \cos 50\pi t \left(\frac{1 + \cos 300\pi t}{2} \right)$

$$= 5 \cos 50\pi t + 5 \cos 50\pi t \cos 300\pi t$$

The maximum frequency component will be $150 + 25 = 175$ Hz.

Thus $f_s = 2 \times 175 = 350$ sample per second.

45. (B) Total sample = $8000 \times 20 = 160$ k sample/sec

Bit for each sample = $7 + 1 = 8$

Bit Rate = $160k \times 8 = 1280 \times 10^3$ bits/sec

46. (B) The sampling rate is $f_s = 44100$ meaning that we take 44100 samples per second. Each sample is quantized using 16 bits so the total number of bits per second is 44100×16 . For a music piece of duration 50 min = 3000- sec the resulting number of bits per channel (left and right) is $44100 \times 16 \times 3000 = 2.1168 \times 10^9$ and the overall number of bits is $2.1168 \times 10^9 \times 2 = 4.2336 \times 10^9$

47. (C) Nyquist Rate = $2 \times 4k = 8$ kHz

Total sample = $4 \times 8 = 32$ k sample/sec

$256 = 2^8$, so that 8 bits are required

Bit Rate = $32k \times 8 = 256$ kbps

48. (D) Nyquist Rate = $2 \times 30k = 60$ kHz

$2^n \geq 6$ Thus $n = 3$, Bit Rate = $60 \times 3 = 18$ kHz

49. (C) Nyquist rate = $2 \times 4 = 8$ kHz

$2^n = 16$ or $n = 4$, Bit Rate = $4 \times 8 = 32$ kbits/sec

50. (C) $f_s = 8$ kHz, $2^n = 256 \Rightarrow n = 8$

Bit Rate = $8 \times 8k = 64$ kb/x

51. (D) $\frac{S_o}{N_o} \propto 2^{2n}$, If PCM is increased from n to $n + 1$,

the ratio will increase by a factor 4. Which is independent of n .

52. (C) If $L = 2$, then $2 = 2^n$ or $n = 1$ ND If $L = 8$, then $8 = 2^n$ or $n = 3$. So relative bandwidth will be tripled.

53. (B) The minimum bandwidth requirement for transmission of a binary PCM signal is $BW = vW$. Since $v = 10$, we have $BW = 10 W$
